



Unraveling the impact of micro- and nano-sized polymethyl methacrylate on gut microbiota and liver lipid metabolism: Insights from oral exposure studies[☆]

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ABSTRACT

Microplastics, particularly polymethyl methacrylate (PMMA), have emerged as significant environmental pollutants, with growing concerns about their impact on various biological processes. However, the effects of chronic PMMA exposure on hepatic lipid metabolism remain insufficiently studied. This research aimed to examine the consequences of chronic exposure to PMMA particles of different sizes (100 nm and 2 μ m) on hepatic lipid metabolism in mice. Female C57BL/6J mice were administered PMMA particles in drinking water over an 8-week period, and the effects on intestinal and liver morphology and function were evaluated. Histopathological analyses, gut microbiota profiling, and serum and liver assays were conducted to assess oxidative stress, lipid metabolism-related biomarkers, and liver metabolomics. The results revealed that PMMA particles accumulated in both the liver and colon, causing liver injury characterized by elevated ALT and AST levels. The exposure also induced oxidative stress by inhibiting the NRF2/HO-1 signaling pathway. Furthermore, PMMA exposure resulted in significant alterations to the gut microbiota and hepatic metabolism. These changes were linked to increased microbial diversity, which impacted cholesterol metabolism through the gut-liver axis. Additionally, the activation of the PI3K/AKT/PPAR γ signaling pathway disrupted hepatic lipid metabolism, leading to increased cholesterol synthesis and hepatic lipid accumulation. This study underscores the potential of PMMA to disrupt both hepatic lipid metabolism and gut microbiota composition, suggesting a novel mechanism by which PMMA exposure could contribute to metabolic disorders and liver disease.

1. Introduction

Plastics, due to their lightweight properties, corrosion resistance, excellent insulation, durability, low cost, and easy availability, have become widely used in industries such as packaging, construction, healthcare, electronics, automotive, and consumer goods. However,

inadequate waste management and recycling practices have led to extensive plastic pollution, contaminating soil, water and air (Deakin et al., 2024). By 2015, approximately 6.3 billion tons of plastic waste were generated, with around 79 % having accumulated in landfills or natural environments. This figure is projected to reach 12 billion tons by 2050 (Geyer et al., 2017). Incinerating plastic waste results in the

Abbreviations: MPs, Microplastics; NPs, nanoplastics; UV, ultraviolet; PMMA, polymethyl methacrylate; As, arsenic; SOD, superoxide dismutase; CAT, catalase; PS, polystyrene; PMMA-NPs, PMMA nanoparticles; MDA, malondialdehyde; TEM, transmission electron microscope; AST, aspartate aminotransferase; ALT, alanine aminotransferase; TC, total cholesterol; TG, triglycerides; LDL-C, low-density lipoprotein cholesterol; HDL-C, high-density lipoprotein cholesterol; LefSe, Linear Discriminant Analysis Effect Size.

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emission of harmful substances, further contributing to environmental pollution. Once in the environment, plastic waste degrades into microplastics (MPs) and nanoplastics (NPs) through ultraviolet (UV) radiation, mechanical abrasion, and biodegradation (Rhodes, 2018). MPs are generally defined as particles smaller than 5 μm , while NPs are typically less than 100 nm in size (Law, 2017). Growing evidence suggests that M-NPs can gather in several human organs, such as lungs, liver, intestines, kidneys, placenta, testes, and blood, raising concerns about potential health risks (Zhu et al., 2024; Zhu et al., 2023; Massardo et al., 2024; Yang et al., 2024; Li et al., 2024a).

Polymethyl methacrylate (PMMA), commonly known as plexiglass or methyl acrylate, is a synthetic polymer valued for its optical transparency, mechanical strength, and durability, making it extensively used in industrial manufacturing, medical devices, optics, and other fields (Ahn et al., 2024; Fang et al., 2024; Wang et al., 2024a). However, its potential biotoxicity warrants attention. Research shows that PMMA exposure significantly inhibits the germination index of rapeseed, with nanoparticles exhibiting greater toxicity compared to microparticles (Dong et al., 2022a). In co-contamination scenarios, PMMA and arsenic (As) display synergistic toxicity, affecting plant growth, lipase and α -amylase activities, and reducing antioxidant enzymes such as superoxide dismutase (SOD) and catalase (CAT) (Ozfidan-Konakci et al., 2023). Additional experiments revealed that exposure to polystyrene (PS) and PMMA nanoplastics in the presence of arsenate alters plant nutrient and hormone levels. Although some mitigation of photosynthetic damage was observed, the antioxidant system was insufficient to fully counteract arsenic toxicity (Dong et al., 2022b). Root and leaf exposure to PMMA microplastics caused more pronounced physiological damage in roots and increased arsenic accumulation. Interestingly, PMMA stimulated root growth in onions, highlighting the broad ecological impacts of plastic pollution on plant health and ecosystems (Biba et al., 2023).

Similar toxicological effects of PMMA have been observed in animal studies. PMMA nanoparticles (PMMA-NPs) have been shown to reduce cholinesterase activity and increase antioxidant defenses in *Hediste diversicolor*, with juveniles displaying heightened sensitivity, including increased biochemical toxicity, disrupted neurotransmission, and altered energy metabolism (Neves et al., 2024; Silva et al., 2023). PMMA also exhibits embryotoxic and reproductive toxicity. PMMA microparticles of varying sizes and concentrations have induced developmental defects, cytogenetic abnormalities, and impaired sperm fertilization in sea urchin embryos and larvae, along with increased mitotic abnormalities and embryo toxicity. Notably, sperm exposure to PMMA has caused significant damage to offspring (Trifuoggi et al., 2019). In zebrafish embryos and larvae, PMMA-NPs have been linked to higher mortality and pericardial edema (Manuel et al., 2022). Additionally, PMMA particles have led to oxidative stress within the gastrointestinal system of *Littorina brevicula*, indicated by higher concentrations of malondialdehyde (MDA), protein carbonyls, and increased DNA damage in digestive gland cells (Chelomin et al., 2023).

Although the effects of PMMA on lipid metabolism are still not fully understood, existing data suggest that PMMA may influence lipid metabolism through its toxicological and biochemical actions. PMMA exhibits complex effects on the physiological states of both plants and animals, including disruptions in antioxidant defenses and cellular integrity, which may indirectly affect lipid metabolic processes, and its own physical properties. Adsorption of adsorbent microplastic particles onto lipid membranes significantly elevates membrane tension, and the cumulative effect of continued exposure leads to an overall decrease in membrane area and an increase in membrane tension (Fleury and Baulin, 2021). Once the lipid membrane is affected by mechanical stress, the fluidity and stability are altered, which may lead to conformational changes in functional proteins such as receptors and enzymes related to lipid metabolism on the membrane, and consequently affecting their ability to bind to the corresponding substrates or signaling molecules (Ren et al., 2024). In addition, PMMA may affect lipid metabolism by

interfering with intracellular signaling pathways. PMMA exposure decreased mitochondrial membrane potential, leading to intracellular ROS accumulation and oxidative stress, and inhibited the AKT/mTOR signaling pathway and affected metabolic pathways such as fatty acids and nucleotides (Xuan et al., 2024). Studies have shown that exposure of *Sparus aurata* to PMMA detected elevated expression levels of key genes related to lipid metabolism (e.g. apolipoprotein A1 and retinoid X receptor) and an overall increase in plasma TG and TC (Brandts et al., 2021). In addition the effect of size on the metabolic toxicity of PMMA is significant. Nanosized particles cause more severe effects on cell membrane potential, ATPase and cholinesterase activities (Zhao et al., 2023; Silva et al., 2023). PMMA-induced oxidative stress may be a key factor influencing lipid metabolism. When the organism is exposed to PMMA, free radicals such as reactive oxygen species will be generated, and these free radicals will attack lipid components such as phospholipids on the cell membrane, leading to lipid peroxidation (Gao et al., 2024; Li et al., 2024b). The peroxidized lipid products may further interfere with the activity of intracellular lipid metabolism-related enzymes, such as fatty acid synthase, lipase, etc., causing an imbalance between lipid synthesis and catabolism. Despite existing studies suggesting that PMMA may affect lipid metabolism through multiple mechanisms, several gaps and controversies remain. For instance, it is unclear whether the lipotoxic effects of PMMA are significantly influenced by particle size, exposure concentration, or duration. Additionally, the long-term effects of PMMA on lipid metabolism and its potential chronic toxicity are still underexplored. In this study, we developed a long-term oral exposure mouse model to examine the impact of PMMA particles of different sizes (100 nm and 2 μm , referred to as PMMA-NP and PMMA-MP, respectively) on hepatic lipid metabolism. The exposure concentration was set at 200 $\mu\text{g/L}$ in reference to previous study, in the normal range of environmental exposures (Lv et al., 2023; Mohamed Nor et al., 2021). The objective was to elucidate the mechanisms underlying PMMA-induced hepatic lipotoxicity. Our findings provide new insights into the potential risks of PMMA exposure in mammals.

2. Materials and methods

2.1. PMMA characterization

Suspensions of PMMA particles, with sizes of 100 nm and 2 μm , were purchased from Wuxi Rigor Biotech (China), and their morphology was analyzed using a transmission electron microscope (TEM, Hitachi, Japan). The PMMA particles were systematically characterized with a Renishaw inVia™ confocal microscope, followed by Raman spectral analysis. The raw spectra underwent polynomial baseline correction and vector normalization to minimize noise and enhance spectral quality. The processed Raman spectra were then compared to reference spectra from a custom library of common microplastic polymers to confirm the identity of the plastic particles.

2.2. Animal treatment

Six-week-old female C57BL/6J mice were obtained from Changsha Tianqin Biotechnology Co. in Changsha, China. With free access to food and water, the animals were housed at 23–25 °C and 50–60 % humidity under a 12-h light/dark schedule. The Institutional Animal Care and Use Committee at Guilin Medical University approved all animal experiments (Approval No. GLMC202303207).

After a one-week acclimatization period, the mice were randomly divided into three groups ($n = 6$). One group served as the control, while the other two groups were exposed to PMMA particles of 100 nm and 2 μm , referred to as PMMA-NP and PMMA-MP, respectively. All groups were fed a standard diet. The PMMA-NP and PMMA-MP groups were fed a 200 $\mu\text{g/L}$ PMMA suspension prepared with ultrapure sterile water as an environmental drinking concentration (Zhang et al., 2022; Shi et al., 2024). The PMMA suspensions were sonicated in advance to ensure

uniform distribution in drinking water (Liu et al., 2022; Zhai et al., 2024). The control group was given regular drinking water. On day 60, following an overnight fast, the mice were euthanized, and tissue samples, intestinal contents, and serum were collected for subsequent analysis.

2.3. Histopathological analysis

To produce 4 μm sections, liver tissues were fixed in 4 % paraformaldehyde and embedded in paraffin. The sections were deparaffinized in xylene, followed by an alcohol gradient and distilled water. Hematoxylin staining was performed for 5 min, followed by rinsing and eosin staining for 3 min. The sections were then dehydrated and sealed with neutral resin. Microscopic observation was carried out, and images were captured using an image acquisition system.

2.4. Liver biochemical and metabolic analysis

The collected blood samples were centrifuged at $1300\times g$ for 15 min to separate the serum. The activities of aspartate aminotransferase (AST) and alanine aminotransferase (ALT) were measured using enzyme colorimetric assay kits (Jiancheng Bioengineering Institute, China; Cat. No. C010-2-1 and C009-2-1, respectively). Levels of MDA, SOD, total cholesterol (TC), triglycerides (TG), low-density lipoprotein cholesterol (LDL-C), and high-density lipoprotein cholesterol (HDL-C) were assessed using kits from Solarbio Science & Technology Co. (China; Cat. No. BC0200, BC0025, BC5165, A111-2-1, A110-2-1, A113-2-1, and A112-2-1, respectively).

2.5. Intestinal flora 16S rRNA sequencing

After 8 weeks of PMMA exposure, cecal contents were collected from each group of mice. Microbial sequencing and data analysis were performed by Wuhan Metviare Biotechnology Co., Ltd. Microbial genomic DNA was extracted from freeze-dried cecal content samples. Universal primers with adapter sequences were used to amplify the V3–V4 regions of the 16S rRNA gene. The library was constructed using the NEB Next® Ultra DNA Library Prep Kit, and its quality and concentration were assessed using an Agilent 5400 and quantified by q-PCR. After passing quality control, the library was sequenced on a NovaSeq 6000 platform (Huang et al., 2023; Xie et al., 2022).

2.6. Liver non-targeted metabolomics

Weighed 20 mg of liver tissue and mixed it with 600 μL of ice-cold 67 % methanol, stored the mixture at 4 °C. Homogenized the sample three times using an ultrasonic homogenizer, the sample was centrifuged at 12,000 rpm for 10 min at 4 °C. Collected the clear supernatant and incubated it at -20°C for 30 min. After incubation, centrifuged at 12,000 rpm and 4 °C for 10 min. Transferred 200 μL of the supernatant to a 5 mm NMR tube for analysis. Converted the raw mass spectrometry data into mzXML format using ProteoWizard, and performed peak extraction, alignment, and retention time correction with XCMS. Peaks with more than 50 % missing values were filtered out, blank values were imputed using KNN, and peak areas were corrected using SVR. Substances with an identification score greater than 0.5 and a QC sample CV of less than 0.3 were extracted. Finally, positive and negative modes were merged, retaining substances with the highest qualitative level and the smallest CV values.

2.7. Real-time quantitative PCR

According to the manufacturer's protocol, total RNAs was isolated from mouse liver tissue using Trizol reagent (Cat. No. 15596026CN, Invitrogen, USA). The RNAs was eluted with pre-warmed RNase-free water. RNAs purity was assessed using a NanoDrop 2000

spectrophotometer (Thermo Fisher Scientific, USA), and high-quality RNAs with an A260/280 ratio between 1.8 and 2.0 was used for further processing. Total RNA was reverse transcribed into cDNA using the RevertAid First Strand cDNA Synthesis Kit (Cat. No. K1622, Thermo Fisher Scientific, USA) following the manufacturer's instructions. mRNA levels were normalized to β -actin and quantified using the $2^{-\Delta\Delta\text{Ct}}$ method.

2.8. Western blotting analysis

Western blot analysis was performed to assess the expression of specific proteins in mouse liver tissue. Total liver proteins were extracted using RIPA lysis buffer. Protein concentrations were determined with a BCA protein assay kit (Solarbio Science & Technology Co., China). Using 7.5 % or 10 % SDS-PAGE gels, equal protein quantities were separated and then moved to a polyvinylidene fluoride (PVDF, 0.45 μm) membrane. The membrane was blocked with 10 % non-fat milk for 1 h, then was incubated overnight at 4 °C with specific primary antibodies. The primary antibodies used were rabbit anti-AKT (AF6261, 1:1000, affinity), rabbit anti-PI3K (AF4369, 1:1000, affinity), rabbit anti-PPAR γ (AF6284, 1:1000, affinity), rabbit anti-NRF2 (16396-1-AP, 1:2000, Proteintech), rabbit anti-HO-1 (10701-1-AP, 1:1000, Proteintech), mouse anti- β -actin (66009-1-Ig, 1:1000, Proteintech). After three washes with Tris-buffered saline containing Tween 20, the membrane was incubated with secondary antibodies for 60 min at room temperature. Secondary antibodies used were Goat Anti-Rabbit IgG(H + L) (SA00001-2, 1:10000, Proteintech), Goat Anti-Mouse IgG(H + L) (SA00001-1, 1:10000, Proteintech). Protein bands were visualized using a gel imaging system (Bio-Rad, USA) and were quantified using ImageJ software.

2.9. Immunofluorescence

Tissue samples are dewaxed, antigenically repaired, and then exposed to 0.1 % Triton X-100 for 5 min. Goat serum solution is added slowly, drop by drop, over a period of 1 h. After exclusion of the inclusion solution, primary antibody against ZO-1 (1:100, Affinity, China) was added and incubated at 4 °C overnight. After washing the primary antibody with PBS, a red fluorescent marker was applied.

2.10. Statistical analysis

Every experiment was performed thrice independently and the results were similar and expressed as mean $\pm$ SD (standard deviation, $n = 3$). Statistical analysis was performed using GraphPad Prism Version 8.0 software. T-test and one - way ANOVA were used for data analysis. "*" represents a P - value < 0.05, "***" represents a P - value < 0.01. A P - value < 0.05 indicates that the difference in the results is statistically significant.

3. Results

3.1. Morphologies and exposure of PMMA in mouse liver and colon

TEM and Raman spectral analysis confirmed that PMMA particles are spherical and exhibit uniform size distribution (Fig. 1A and B). Moreover, PMMA particles were identified in both liver and colon tissue sections (Fig. 1C). These results suggest that after 8 weeks of exposure, PMMA had particles accumulated in the liver and colon of mice.

3.2. PMMA induced multi-organ damage

After 8 weeks of exposure, diet and water intake, body weight, colon length, and liver coefficient were recorded. PMMA-NP group exhibited a significant increase on body weight and liver index, and a decrease on colon length (Fig. 2A–F). PMMA exposures resulted in significantly

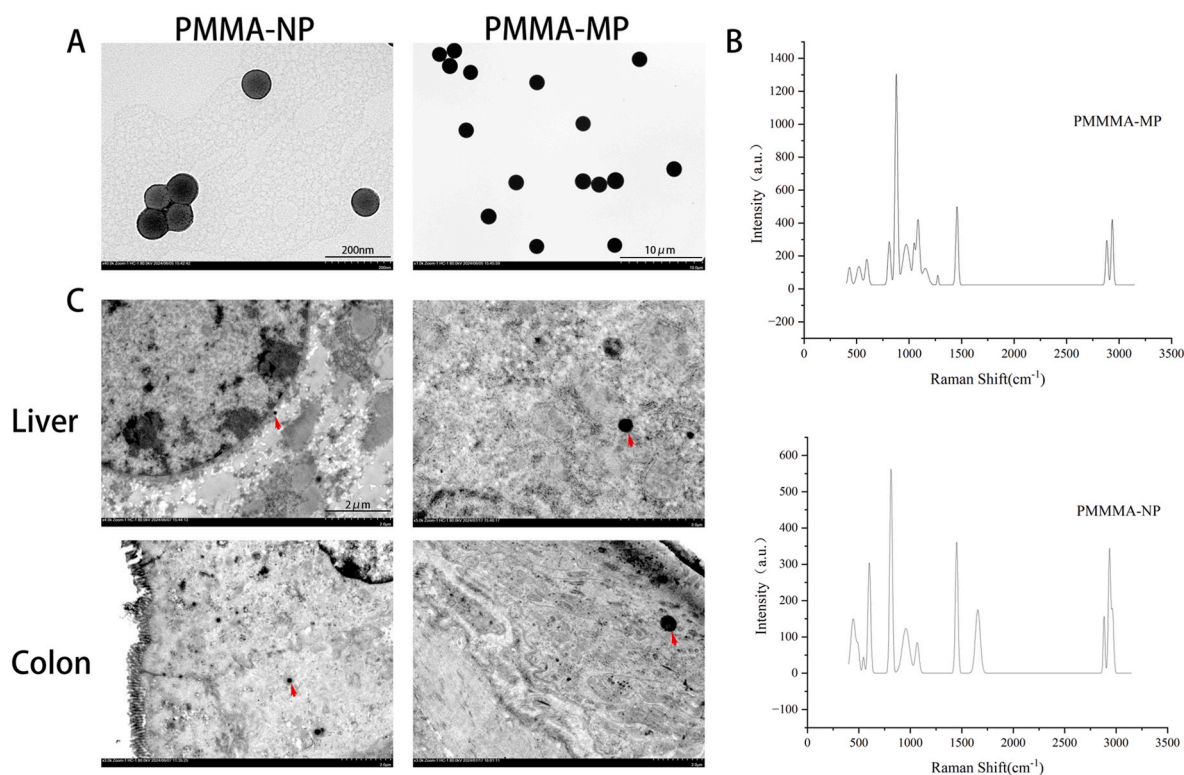


Fig. 1. Characterization of with different particle sizes. (A) TEM micrographs. (B) Processed Raman spectra confirming the identity of PMMA. (C) PMMA particles of varying sizes observed in mouse liver and colon tissues, with red arrows indicating PMMA particle localization. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

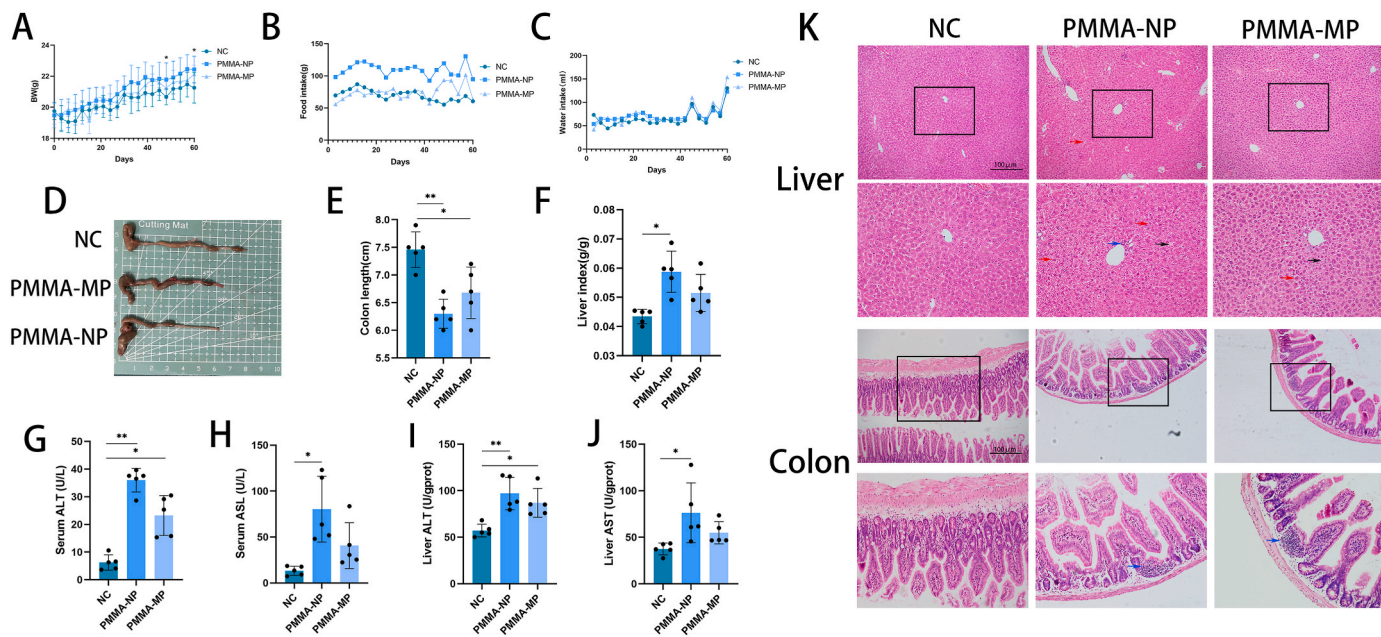


Fig. 2. Organ damage after PMMA exposure. (A) Mouse body weight changes (n = 6). (B) Mouse food intake (n = 6). (C) Mouse water intake (n = 6). (D–E) Colon length (n = 5). (F) Mouse liver organ coefficient (n = 5). (G–J) Mouse serum and liver ALT and AST levels (n = 5). (K) HE staining images of mouse liver, colon (n = 3). Red arrows indicate steatosis, blue arrows indicate inflammatory cell infiltration, and black arrows indicate cellular edema. Statistical significance is indicated by *: $P < 0.05$ and **: $P < 0.01$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

elevated ALT and AST levels in the liver and serum, with PMMA-NP being more significant, which indicating liver damage (Fig. 2G–J). Histopathological evaluation using HE staining revealed severe liver, intestinal, and renal damage. Compared to the control group, PMMA-NP

group displayed more severe hepatocellular steatosis, inflammatory cell infiltration, blurred liver lobules, vacuolation, and uneven hepatic cords with mild disruption (Fig. 2K). Intestinal damage was marked by thinning of the basal layer, epithelial destruction, lymphocyte aggregation,

and inflammatory cell infiltration (Fig. 2K).

3.3. PMMA affects gut microbial composition and intestinal barrier function

To investigate the effect of PMMA on gut microbiota, we performed 16S rRNA sequencing on the fecal samples from mice. Analysis revealed changes in β -diversity in the PMMA groups (Fig. 4A). In Fig. 3B, PCA results indicated that PMMA exposure altered gut microbiota composition. The Venn diagram of ASV/OTU counts (Fig. 3C) showed a significant increase in microbiota diversity in PMMA-NP. The bar chart of species relative abundance showed top 10 species for each group of abundance at phylum level (Fig. 3D). The NC group exhibited a higher abundance of Bacteroidetes compared to the PMMA groups, whereas the Firmicutes abundance was notably lower in the PMMA groups (Fig. 3D). Consequently, the Bacteroidetes/Firmicutes ratios were reduced in both PMMA groups, with a significant decrease in the PMMA-NP group ($P < 0.05$) (Fig. 3E). In addition, Verrucomicrobia was more abundant in the PMMA-MP group compared to both NC and PMMA-NP, while Actinobacteria was relatively more abundant in the NC group. LefSe analysis identified colonies with significant differences in abundance (Fig. 3F), indicating the extent of variation among subgroups. In addition, we examined the expression of mouse intestinal tight junction protein ZO-1. Compared with the NC group, ZO-1 expression was decreased in all experimental groups (Fig. 3G), indicating the damage to the intestinal barrier by PMMA exposure.

3.4. PMMA suppresses NRF2/HO-1 signaling pathway and induces oxidative stress in mouse liver

TEM images revealed that mitochondrial cristae in liver cells were clearly visible in the NC group. In contrast, the PMMA-NP group showed closely packed mitochondria and disappearance of cristae, while the PMMA-MP group exhibited mitochondrial and lysosomal membrane rupture, indicating mitochondrial oxidative stress damage in the liver (Fig. 4A). To investigate the effect of the induced oxidative stress in PMMA-induced liver injury, liver and serum levels of MDA and SOD, as well as liver expression of NRF2 and HO-1, were assessed. PMMA-NP exposure significantly increased liver and serum MDA levels while decreasing SOD activity (Fig. 4B–E). Additionally, Western blot analysis demonstrated a significant reduction in NRF2 and HO-1 expression (Fig. 4F–H).

3.5. PMMA-induced changes in liver metabolomics

We performed untargeted metabolomics analysis on mouse liver tissue to investigate the metabolic disturbance caused by PMMA exposure. PCA showed the difference in hepatic metabolic profile between the PMMA-NP and NC, while the PMMA-MP exhibited moderate variations (Fig. 5A). OPLS-DA further demonstrated the separation of the PMMA-NP group from the NC group (Fig. 5B). These findings indicate that PMMA exposure disrupts liver metabolism. Comparing the metabolite profile with NC group identified 140 upregulated and 127 downregulated metabolites in PMMA-NP (Fig. 5C). The bar graph of differential metabolites illustrated the changes in metabolite levels in the PMMA-NP group compared to the NC group (Fig. 5D). The red bars represented up-regulated metabolites, including Geranylgeraniol, Asclepin, and various phospholipids, indicated that PMMA may influence cell signaling and membrane structure by enhancing lipid metabolism and inflammatory responses. The green bars indicated down-regulated metabolites such as Acrylic acid, Spermidine, and 1,2-Dioleoyl-sn-Glycero-3-Phosphocholine, which are associated with lipid metabolism, cell proliferation, and autophagy function, indicated that PMMA nanoparticles may inhibit these biological processes.

KEGG pathway enrichment analysis identified significant enrichment in several metabolic pathways in PMMA-NP group, including

linoleic acid metabolism, alpha-linolenic acid metabolism, choline metabolism, fatty acids metabolism. The metabolites in these pathways were significantly altered in the PMMA-NP group, suggesting that PMMA treatment affects lipid metabolism. In addition, key lipid metabolism pathways such as glycerophospholipid metabolism and biosynthesis of unsaturated fatty acids were significantly enriched, indicating extensive interference with lipid synthesis and catabolism by PMMA-NP. Differences were also observed in pathways such as steroid hormone biosynthesis and chemical carcinogen receptor activation, which could indicate potential toxic effects or inflammatory responses due to PMMA exposure (Fig. 5F). To further investigate the relationship between differential metabolites and the microbiota, we performed Spearman correlation analysis between the differential metabolites and gut microbiota. Fig. 5G shows the correlation between different compounds and microbial families. The Lactobacillaceae family and Peptococcaceae family of the phylum Firmicutes showed positive correlation with metabolites such as Zerumbone, Hydroxy-beta-cyclocitra, Carnitine C18:3 and others. correlation; and a positive correlation with Carnitine C18:2, 1-Benzylpiperazine, and LPS(22:6) metabolites were negatively correlated.

3.6. Effect of PMMA on hepatic lipid metabolism

After 8 weeks of PMMA exposure, histopathological changes in liver tissue were observed. TEM images revealed numerous vacuoles in the cytoplasmic matrix of liver tissue in both the PMMA-NP and PMMA-MP groups, with lipid droplets also detected in the PMMA-NP, providing evidence for PMMA-induced liver lipid deposition (Fig. 6A).

To further investigate whether PMMA affects lipid metabolism in mice, we measured serum levels of HDL-C, LDL-C, TC, and TG. The results indicated that PMMA exposure led to alterations in these serum levels, and the alterations in PMMA-NP were more significant. Specifically, LDL-C and TC levels were significantly increased, while HDL-C levels were significantly decreased. Although TG levels did not show a significant change, there was a noticeable upward trend (Fig. 6B–E).

qPCR results indicated that PMMA exposure downregulates genes involved in cholesterol catabolism, such as LPL, Abca1, Abcg1 and PPAR α (Fig. 6F–I). Additionally, genes associated with cholesterol synthesis and reverse transport, including Hmgcr, Hmgcs1, PCSK9, were also downregulated (Fig. 6J–L), with these changes being more severe in PMMA-NP compared with NC and PMMA-MP. These findings suggested that PMMA promotes increased cholesterol synthesis in the liver and cholesterol deposition, leading to disrupted liver lipid metabolism.

3.7. PMMA activates the PI3K/AKT/PPAR γ pathway to induce liver lipid metabolism disorders

The PI3K/AKT/PPAR γ pathway is essential for regulating lipid metabolism. Activation of PI3K triggered an increase in AKT expression while inhibiting PPAR γ expression, which promoted lipid and cholesterol synthesis, leading to their accumulation in the liver (Lee et al., 2015). qPCR and Western blot analyses demonstrated that PMMA-NP exposure led to a more significant upregulation of PI3K/AKT and downregulation of PPAR γ compared to PMMA-MP, contributing to more severe disturbances in hepatic lipid metabolism (Fig. 7A–H).

4. Discussion

Among emerging pollutants, M-NPs have drawn considerable attention due to their capacity to enter human and animal systems and adversely impact health (Brander et al., 2024). Compared to MPs, NPs exhibit smaller particle sizes, enabling greater cellular membrane permeability and consequently inducing more severe biological toxicity (Lu et al., 2024).

The liver and intestines are key target organs for M-NPs, with exposure leading to notable lipid metabolism disturbances (Fan et al.,

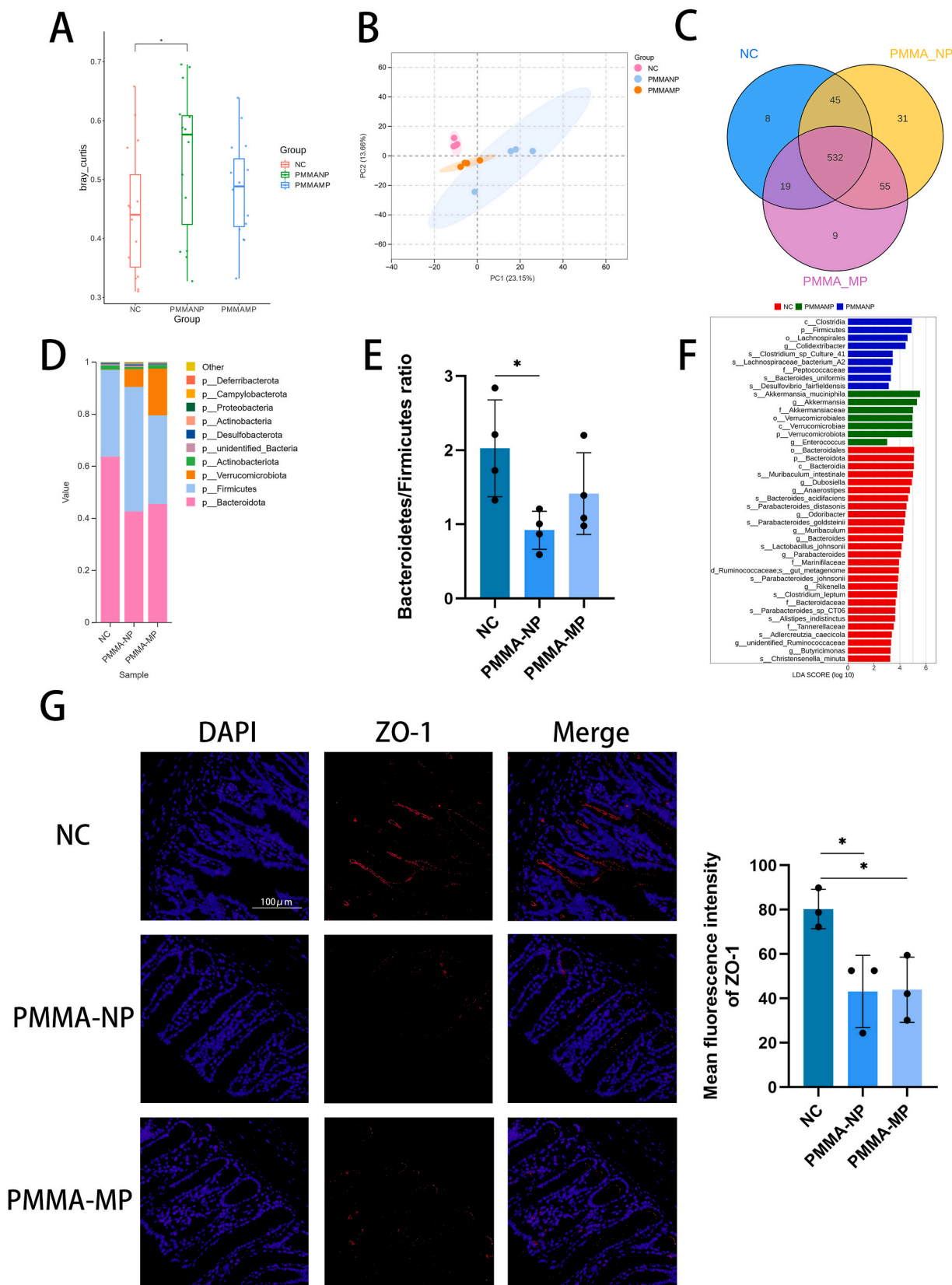


Fig. 3. Disruption of gut microbiota in mice by PMMA (A) Boxplot of β -diversity of gut microbiota, based on non-parametric Kruskal-Wallis test. (B) Principal Component Analysis (PCA) illustrating multivariate statistical analysis. (C) Venn diagram showing ASV/OTU abundance histograms at the phylum level across different regions. (D) Bar graph of species relative abundance at the phylum level, displaying the top 10 species with the highest abundance. (E) Bacteroidetes/Firmicutes ratio (F) Linear Discriminant Analysis Effect Size (LEfSe) analysis, based on non-parametric Kruskal-Wallis test, highlighting phyla with LDA scores ≥ 3 and differential abundance of gut microbiota among the three groups. (G) expression of ZO-1 in the colon tissues of mice in each group (tissue immunofluorescence). Statistical significance is indicated by *: $P < 0.05$.

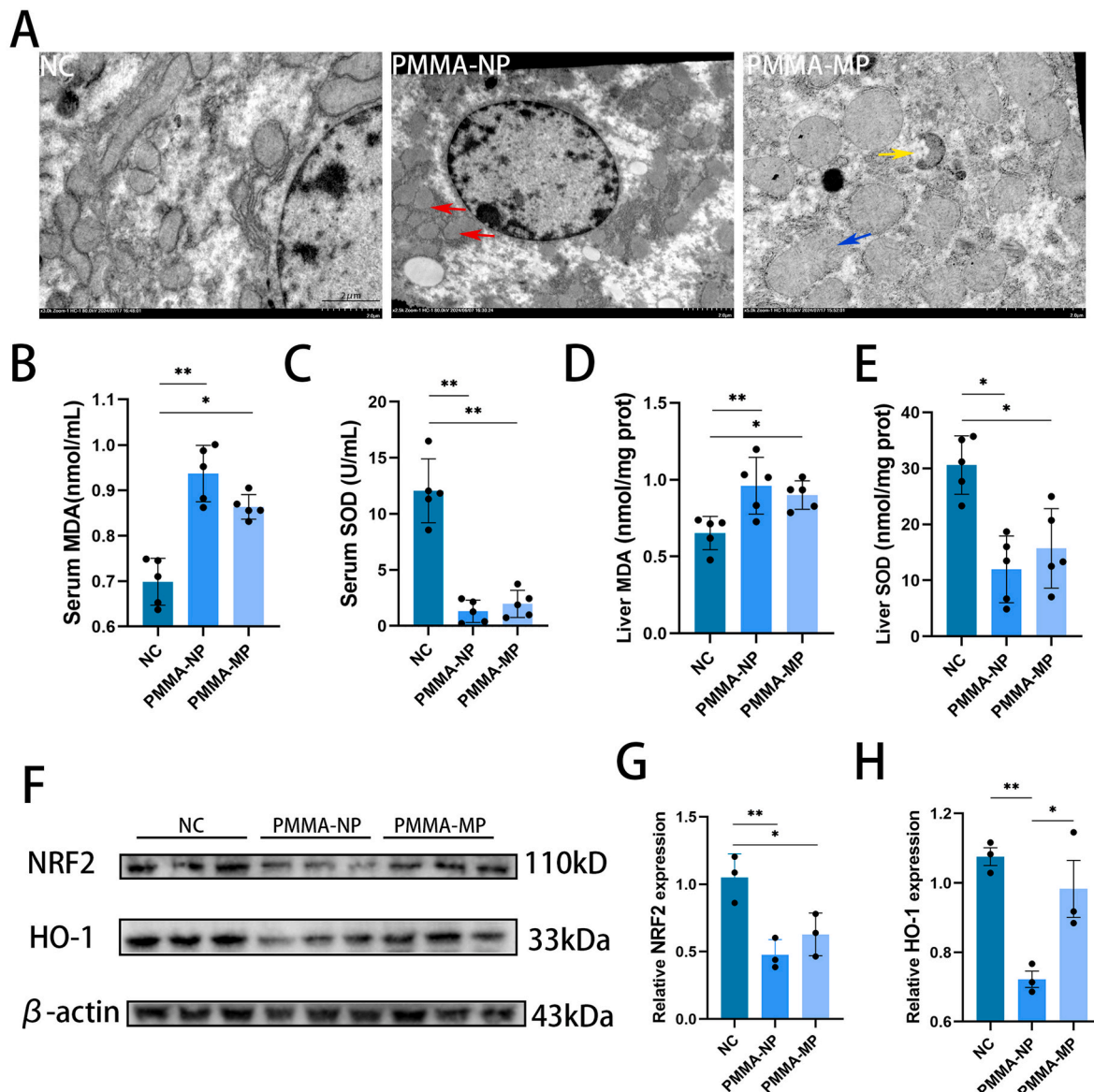


Fig. 4. PMMA-Induced Oxidative Stress through Suppression of Antioxidant Pathways. (A) TEM images of mouse liver; red arrows indicate loss of mitochondrial cristae, blue arrows denote incomplete mitochondrial membrane rupture, and yellow arrows highlight lysosomal rupture. (n = 3) (B) MDA levels in mouse serum (n = 5). (C) SOD levels in mouse serum (n = 5). (D) MDA levels in mouse liver (n = 5). (E) SOD levels in mouse liver (n = 5). (F–H) Protein expression of NRF2 and HO-1 in mouse liver. (n = 3). Statistical significance is indicated by *: $P < 0.05$ and **: $P < 0.01$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2024; Ma et al., 2024). However, the specific mechanisms through which M-NPs influence the gut-liver axis and hepatic cholesterol metabolism remain unclear. PMMA, a plastic extensively utilized in optics, construction, and medical applications, has been scarcely studied in terms of its potential risks and the mechanisms through which it affects liver cholesterol metabolism.

In the present study, we employed PMMA particles with diameters of 100 nm (PMMA-NP) and 2 μ m (PMMA-MP) to assess their hepatotoxicity in mice. TEM confirmed the accumulation of PMMA particles in the liver and colon of exposed mice, with both sizes of PMMA being detectable.

Following NPs exposure, significant impairment of hepatic antioxidant capacity was observed, leading to oxidative stress (He et al., 2024). Additionally, MPs have been implicated as potential inducers of non-alcoholic fatty liver disease (Del Piano et al., 2024). Levels of ALT and AST are commonly utilized to evaluate liver function (Dong et al., 2024a). Our results indicated that ALT and AST levels were significantly

elevated in mice exposed to both PMMA-NP and PMMA-MP, suggesting hepatic injury. Histopathological analysis revealed variable degrees of tissue damage in the liver and colon of PMMA-exposed mice, with smaller particle sizes eliciting more severe damage, consistent with previous findings (Zhang et al., 2023).

Lipid peroxidation byproducts such as MDA and antioxidant enzymes like SOD are well-established markers of oxidative stress (Tao et al., 2024a). NRF2 (nuclear factor erythroid 2-related factor 2) and HO-1 (heme oxygenase-1) are critical in modulating cellular responses to oxidative stress, regulating the expression of genes involved in antioxidant defense and cellular homeostasis (Hassan et al., 2024). Inhibition of the Nrf2/keap1 antioxidant signaling pathway intensifies oxidative stress and promotes lipid peroxidation (Bao et al., 2024), leading to cell membrane damage (Zhang et al., 2024a). In addition, activation of the NRF2 pathway can mitigate oxidative stress and inhibit ferroptosis to achieve anti-atherosclerotic effects (Wang et al., 2024b). Based on previous studies, co-exposure to polystyrene nanoparticles and

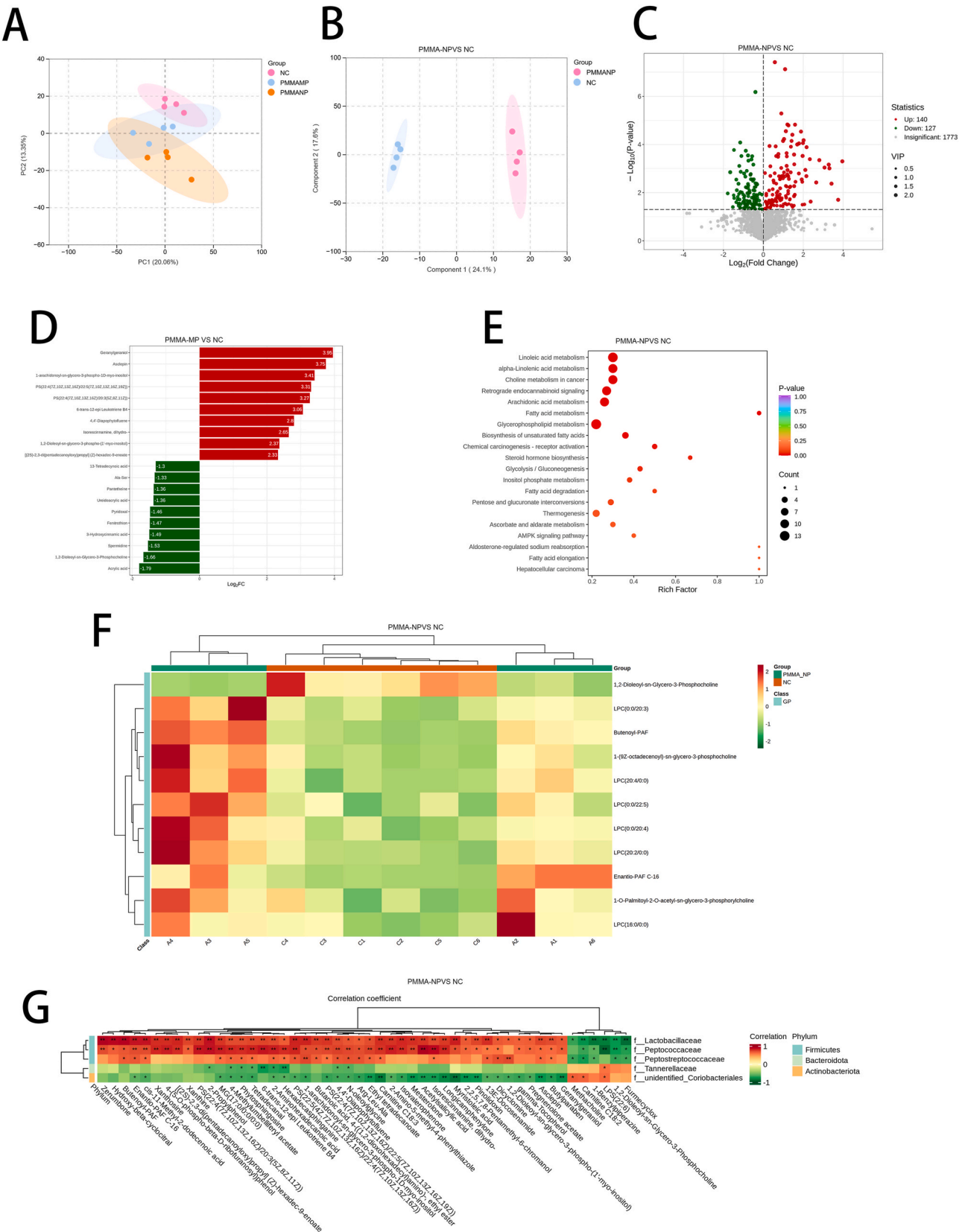


Fig. 5. Disruption of Mouse Liver Metabolism Following PMMA Exposure (A) PCA of overall metabolomics samples. (B) OPLS-DA of PMMA-NP vs NC (C) Volcano plot of PMMA-NP vs NC. (D) Bar chart of differential metabolites in PMMA-NP vs NC. (E) Cluster heatmap of KEGG differential metabolites in PMMA-NP vs NC. (F) KEGG pathway enrichment bubble plot for PMMA-NP vs NC. (G) Hierarchical clustering heatmap of correlations between differential microorganisms and differential metabolites.

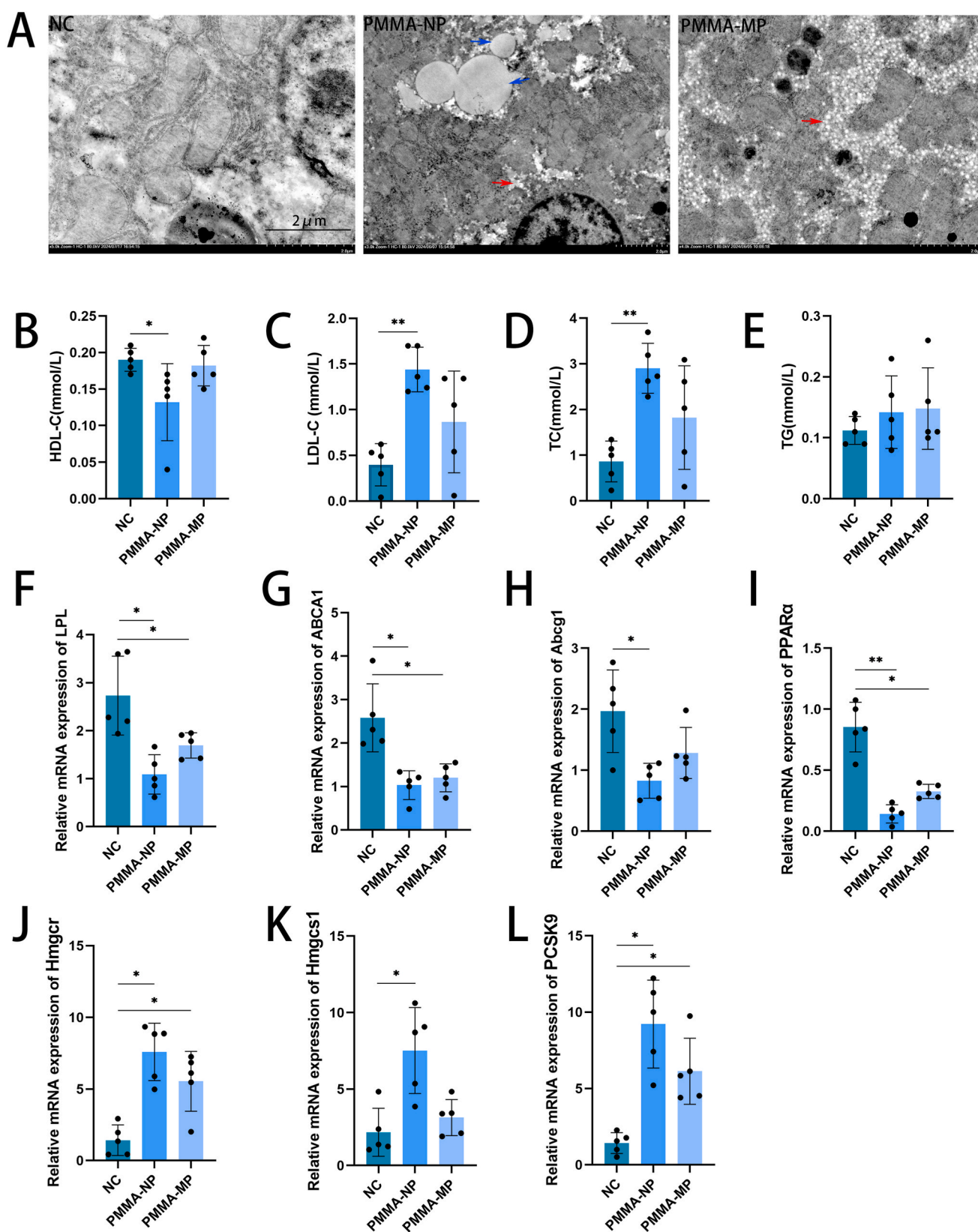


Fig. 6. PMMA exposure induces liver lipid accumulation and disrupts lipid metabolism-related gene expression. (A) TEM images of the liver, with blue arrows indicating lipid droplets and red arrows denoting vacuoles in the cytoplasmic matrix (n = 3). (B–E) Serum levels of HDL-C, LDL-C, TC, and TG in mice (n = 5). (F–L) Expression of cholesterol metabolism-related genes in the liver of mice (n = 5). Statistical significance is indicated by *: P < 0.05 and **: P < 0.01. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

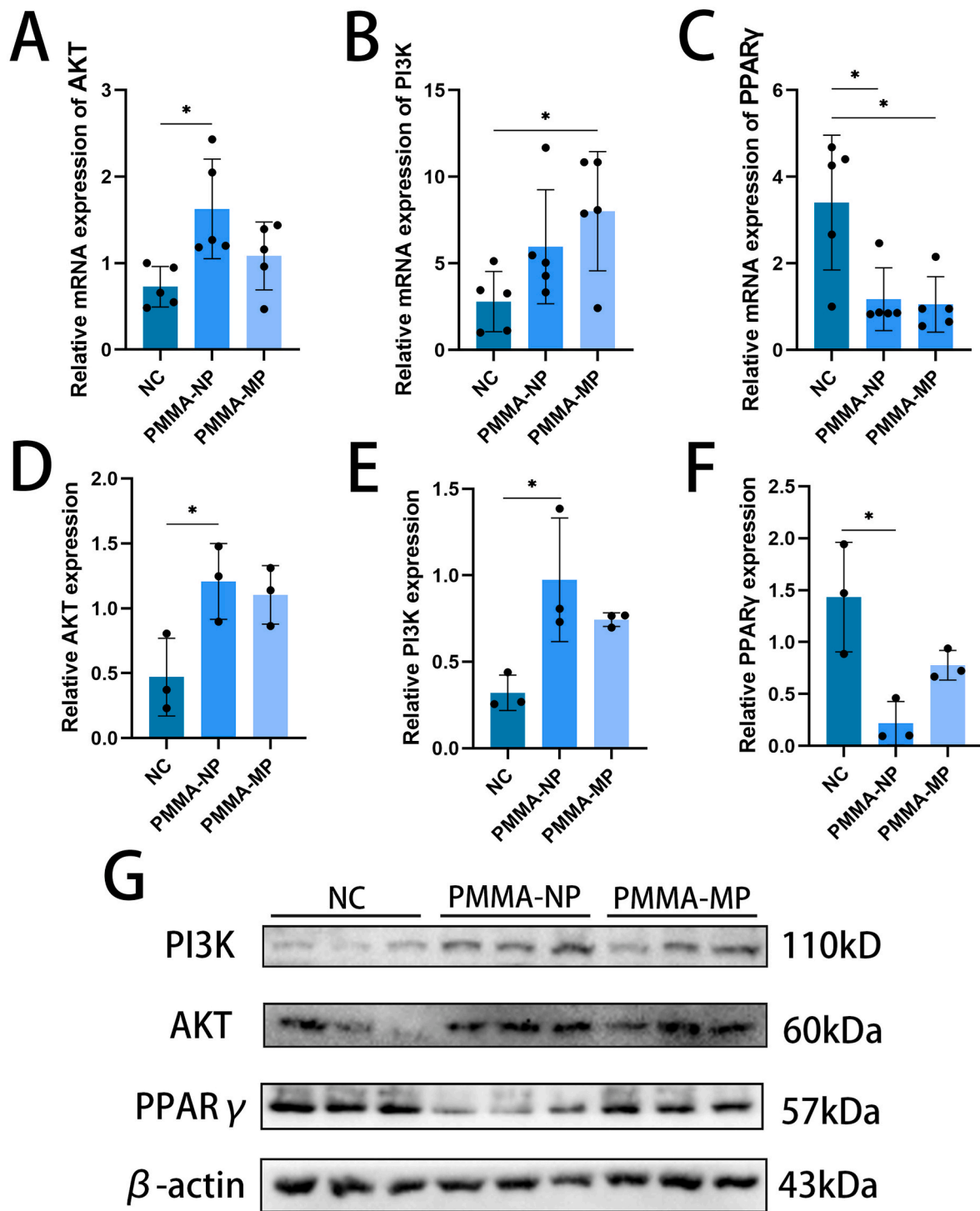


Fig. 7. PMMA activates the PI3K/AKT/PPAR γ signaling pathway (A–C) Gene expression of PI3K, AKT, and PPAR γ in mouse liver (n = 5); (D–G) Protein expression of PI3K, AKT and PPAR γ in mouse liver (n = 3). Statistical significance is indicated by *: P < 0.05 and **: P < 0.01.

As exacerbates oxidative stress in the testes, characterized by increased MDA levels and reduced expression of NRF2 and HO-1 (Rao et al., 2024). Bisphenol S induces oxidative stress by elevating pancreatic reactive oxygen species and inhibiting antioxidant enzyme activity, also reduces fatty acid transport, resulting in lipid deposition in the hepatopancreas (Pu et al., 2024). Mitochondrial oxidative stress is considered as accelerator of organ fibrosis (Moro et al., 2016). It results in damage to mitochondrial structures. Oxidative stress results in mitochondrial swelling, loss of cristae structure, and alterations in oxidative stress markers, including SOD, MDA, as well as changes in the expression of

NRF2 and HO-1 (Moradi Khankani and Hossein, 2024; Tao et al., 2024b). These findings align with our results.

Oxidative stress produces harmful substances, interferes with nutrient metabolism, changes the survival and growth conditions of microorganisms, and ultimately leads to changes in the structure and proportion of the intestinal microbiota (Zhang et al., 2024b). The gut microbiome is highly susceptible to changes induced by factors such as physical activity, diet, pharmacological agents, and environmental exposures, playing a critical role in host health. When mice are exposed to PMMA resulting in oxidative stress, it alters the gut microbiota through

multiple mechanisms (Sun et al., 2024). Disruptions in the gut microbiome are closely associated with increased intestinal permeability and lipid metabolism disorders (Zhang et al., 2024c). Previous studies demonstrated that M-NPs exposure could perturb the gut microbiota composition in mice, zebrafish, and humans (Hong et al., 2024; Junaid et al., 2024; Su et al., 2024). Our findings confirmed that PMMA exposure significantly altered gut microbiota diversity, with notable differences in microbial composition between treatment groups. In the intestinal environment, pH value increases, oxygen content changes, inhibiting acidophilic and anaerobic bacteria, and creating conditions for alkali-resistant bacteria and parthenogenetic anaerobes (Firman et al., 2022). The decreased ratio of Bacteroidetes/Firmicutes increasing cholesterol and fat absorption and storage by influencing the effects of mRNA expression of lipid-metabolizing enzymes and altering bile acid metabolism, thus impairing cholesterol breakdown, potentially promoting obesity, type 2 diabetes mellitus, hypertension, dyslipidemia, and other metabolic disorders (Wu et al., 2023; Wang et al., 2018; Tokarek et al., 2021; Cui et al., 2013). Additionally, a reduction in Actinobacteria and an increase in Verrucomicrobia were linked to elevated levels of atherosclerosis-associated lipid metabolites in mice (Li et al., 2017). Notably, PMMA-MP exposure resulted in more increases in Verrucomicrobia than PMMA-NP exposure, which we hypothesize is due to PMMA-NP's ability to increase intestinal permeability, facilitating the translocation of endotoxins and bacterial metabolites into the bloodstream. In contrast, PMMA-MP, due to its larger size, accumulates in the intestinal lumen, causing more extensive local damage. Changes in gut microbiota induced by PMMA exposure could further modulate liver metabolism through the gut-liver axis, and the gut-liver axis interactions are also perturbed by hepatic lipid metabolism and gut microbiota (Gupta et al., 2020; Shi et al., 2022). High correlation between dysregulated metabolites such as Zerumbone, Carnitine, LPS and important gut microbiota such as Lactobacillaceae family and Peptococcaceae family in the phylum Firmicutes was observed after exposure to PMMA. Reduced expression of the intestinal tight junction protein ZO-1, impaired intestinal barrier, and disruption of mucosal integrity alter the microenvironment for microbial attachment, and survival affect hepatic metabolite levels via the gut-liver axis (Fu et al., 2025). In addition, PMMA may also increase the vulnerability of gut to harmful substance invasion by influencing the physical barrier function (Jin et al., 2019). We hypothesize an association with the fact that PMMA reduces antioxidant capacity in mice and induces hepatic lipid metabolism disorders through the gut-liver axis after disrupting the intestinal barrier (Shaoyong et al., 2023). Significant alterations were observed in overall hepatic metabolite profiles. Bacteroides affects the synthesis of Geranylgeraniol by influencing the metabolic conversion of bile acids, resulting in its up-regulation, indicating potential interference of the mevalonate pathway, which is pivotal for cholesterol biosynthesis (Ho et al., 2018; Chen et al., 2024). Furthermore, Phospholipids such as PS (22:4), 1-arachidonoyl-glycerophospho-inositol, and others were involved in maintaining membrane integrity and fluidity, indirectly affecting cholesterol transport by modulating membrane-associated lipid transport proteins (Vance, 2008). The upregulation of these phospholipids may reflect a compensatory response to oxidative stress, where cells increase lipid production to repair membrane damage and maintain cellular integrity (Tao et al., 2024c). Metabolites of Proteobacteria and Actinobacteria may act as signaling molecules to upregulate the amount of Acrylic acid and pantetheine entering the enterohepatic circulation by altering bile acid secretion (Singh et al., 2023; Martin et al., 2021). Down-regulation of Lactobacillus and 13-tetradecyloic acid reduces PPAR α expression, decreases fatty acid catabolism, and increases fat accumulation (Rivero-Gutierrez et al., 2019; Smarkusz-Zarzecka et al., 2020). These metabolites are critical for fatty acid degradation and cholesterol metabolism, and their reduction may impair these processes, contributing to lipid accumulation in hepatocytes (Evans et al., 2014). 3-hydroxycinnamic acid has multiple associations with the gut-liver axis, it promotes the growth of probiotics such

as bifidobacteria and lactobacilli, maintains the intestinal mucosal barrier function, and also promotes the oxidative metabolism of fatty acids and reduces the deposition of lipids in the liver (Wang et al., 2023). The decreased levels of 3-hydroxycinnamic acid may suggest a reduced antioxidant defense, correlating with oxidative stress-induced lipid metabolism disruption in hepatocytes (Dong et al., 2024b). Cholesterol, a precursor for steroid hormones, shows downregulated biosynthesis, likely due to the altered gut microbial metabolites affecting liver cholesterol pathways (Huang et al., 2025). Linoleic acid and alpha-linolenic acid maintain intestinal microecological balance and intestinal mucosal barrier integrity by promoting the growth of intestinal probiotics and inhibiting harmful bacteria and metabolites (Yuan et al., 2022; Marion-Letellier et al., 2008; Wang and Wang, 2023). Fatty acids, particularly unsaturated fatty acids, may influence cholesterol distribution and transport by modulating PPAR γ activity, which in turn upregulates Abca1 and Abcg1 expression, promoting cholesterol efflux from cells (Yang et al., 2018). The enrichment of inositol phosphate metabolism and pentose and glucuronate interconversions pathways indicates that PMMA exposure may impair cellular energy metabolism and antioxidant defense mechanisms, potentially exacerbating oxidative stress (Xiang et al., 2020). Disruptions in choline metabolism may interfere with lipid transport and storage, given its critical role in phospholipid synthesis and lipoprotein formation (Zhu et al., 2014), while Retrograde endocannabinoid signaling and arachidonic acid metabolism are involved in regulating liver inflammation and lipid homeostasis, and their dysregulation may exacerbate lipid accumulation and hepatic inflammation (Wang et al., 2021). NPs can accumulate in the liver via systemic circulation, affecting lipophagy and inducing lipid accumulation in hepatocytes (Fan et al., 2024). Exposure to PMMA elevated serum levels of LDL-C and TC while reducing HDL-C levels, with no significant change in TG levels. Abca1, involved in HDL particle formation and critical for cholesterol efflux, was downregulated, leading to intracellular cholesterol accumulation (Alruhaimi et al., 2024). HMGCR and HMGCS1, two key enzymes in the mevalonate pathway of cholesterol biosynthesis, are regulated by the PI3K pathway, which stimulates cholesterol synthesis (Zhao et al., 2020). Increased HMGCS1 activity likely enhances the synthesis of cholesterol precursors, contributing to elevated serum cholesterol levels, particularly LDL-C (Heida et al., 2021). Study found that by decreasing the abundance of Lactococcus and other bacteria associated with cholesterol synthesis, can reduce the expression of HMGCS1 to improve the dysregulation of cholesterol metabolism (Wang et al., 2022). Furthermore, activation of the PI3K/AKT pathway promotes the degradation of LDL receptors via interaction with PCSK9, elevating LDL-C levels (Dutka et al., 2024; Yang et al., 2018). Our findings demonstrated significant downregulation of LPL, Abca1 and Abcg1, and upregulation of HMGCR, HMGCS1 and PCSK9 in the liver following PMMA exposure. Activation of the PI3K/AKT pathway inhibited PPAR γ expression, impairing lipid and cholesterol transport, and leading to their accumulation in non-adipose tissues, consistent with our findings (Lee et al., 2015), consistent with our results.

5. Conclusion

As an emerging environmental contaminant, PMMA can accumulate in the intestines, liver and other organs of mammals after prolonged oral exposure and cause pathologic and functional damage, and there is a significant relationship with PMMA particle size. Smaller PMMA particle sizes, due to their higher bioavailability, are more likely to penetrate cell membranes, leading to more pronounced gut microbiota disruption. Changes in the gut microbiota may further triggered disturbances in hepatic metabolomics and increased hepatic oxidative stress. We hypothesize that these factors collectively interfere with cholesterol synthesis and metabolism by activating the hepatic PI3K/AKT/PPAR γ pathway and the expression of genes related to cholesterol metabolism, leading to disorders of hepatic cholesterol metabolism.

CRediT authorship contribution statement

Peng Chen Zheng: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation. **Xin Qiang Pan:** Validation. **Yi Jiong Zhou:** Validation. **Keng Po Lai:** Writing – review & editing, Supervision, Funding acquisition. **Rong Li:** Writing – review & editing, Supervision. **Xiao Xi Zhang:** Writing – review & editing, Resources, Conceptualization.

Ethical approval

Given the retrospective nature of the study, ethical approval was waived by the Ethics Committee of Guilin Medical University and all procedures being performed were part of routine care.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2025.126157>.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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